

Powers and Exponents

Flagship

Lesson 6-1-flagship

Name: _____ **Date:** _____ **Class:** _____

SOUND STUDIO MISSION

The Doubling Mixboard

You are the lead sound engineer at Amplitude Studios. Every time you flip a switch on the new mixboard, the volume doubles — that is the power of exponents. The headline act goes on in one hour, and you must master powers before the speakers can be safely dialed in.



TODAY'S OBJECTIVES

Content Objective: I can write and evaluate numbers in exponent form using a base and a power.

Language Objective: I can explain my work using the words exponent, base, power, and evaluate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Exponent

Base

Power

Evaluate



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The small raised number that shows how many times the base is multiplied is the

2

The number being multiplied repeatedly in a power, like the 3 in 3^4 , is the

3

A number written with a base and an exponent, like 3^4 , is a

4

To find the value of an expression, like $2^3 = 8$, is to it.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

**In the power 2 cubed, which number is the base and which is the exponent?
What job does each one do when you evaluate it?**

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Exponent** — A small number that tells how many times to multiply the number by itself.
Exponente — Un número pequeño que dice cuántas veces multiplicar el número por sí mismo.

☐ **Base** — The number that gets multiplied by itself.
Base — El número que se multiplica por sí mismo.

☐ **Power** — A number written with a base and an exponent, like 2^3 .
Potencia — Un número escrito con una base y un exponente, como 2^3 .

☐ **Evaluate** — To find the value of an expression.
Evaluar — Encontrar el valor de una expresión.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The base is ____ and it is the number being ____.

La base es ____ y es el número que se está ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Each step multiplies by 2 again, so repeated multiplication can be written as a power like 2 cubed.

Evidence: Students see each volume is the previous one times 2 and write $2 \times 2 \times 2$ as 2 cubed, naming 2 as the base and 3 as the exponent.

Reasoning: This evidence connects the definition of exponent to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The base is ____ and it is the number being ____.**

La base es ____ y es el número que se está ____.

- **The exponent is ____ and it tells how many ____.**

El exponente es ____ e indica cuántas ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.